Homework 1: Dictionaries and Supervised Text Classification

Deadline: November 9, 2026

Setup

Question. How well can a word list classify sentiment, and what does a trained classifier learn that the dictionary misses?

Use the same texts, labels, and test split for every method. You may adapt code from class, but explain your choices and report your own analysis. Python and R are both welcome.

Task 1: Data and Research Design

Use `meeting_slides/meeting_05/reviews.csv` in the class repository. Read it with Latin-1 encoding. The columns are `value` (text) and `polarity` (positive or negative). It contains 10,662 Rotten Tomatoes snippets, with 5,331 in each class.

1. **Unit and Labels:** Define the unit of analysis and the target label. Treat *positive* as the positive class throughout. Report the number of observations, missing texts, and class balance.

2. **Training and Test Sets:** Create a stratified 80% training / 20% test split, with random seed 2026. Report its sizes. Set the test data aside until all dictionaries, model settings, and classification thresholds have been fixed.

3. **Label Limitations:** Explain one limitation of these labels: they come from the source review's positive or negative rating, rather than a separate human judgment of each isolated snippet. What might this mean for interpreting an apparent classification error?

Task 2: Dictionary Classification

1. **Dictionary and Preprocessing:** Start with the positive and negative word lists from Class 4, or a published sentiment dictionary that you cite. Specify your tokenization and matching rules. Explain any preprocessing choices; removing stopwords or stemming is not compulsory.

2. **Classification Rule:** Define your score and prediction rule mathematically. For example, if $c_+(d)$ and $c_-(d)$ count positive and negative dictionary matches in text d:

$$s_D(d) = c_+(d) - c_-(d), \qquad \widehat{y}_D(d) = \mathbf{1}\{s_D(d) > 0\}.$$

   Here $\widehat{y} = 1$ means positive. State how your rule handles ties and texts with no matches. Explain whether repeated occurrences count more than once.

3. **Revision:** Inspect training errors and make at least three specific revisions to terms, weights, or matching rules. Explain the evidence for each change. You may include a simple negation rule, but it is not required. Preserve both the original and revised versions.

4. **Training Performance:** Report the original and revised dictionaries' training precision, recall, and F1, plus the share of training texts with no dictionary matches. Explain why improvement on these texts is not yet evidence of better performance on new texts.

Task 3: Supervised Classification

1. **Model Choice:** Choose **one** model: multinomial Naive Bayes, logistic regression, or a linear SVM. Explain your choice and text representation. Naive Bayes requires nonnegative features.

2. **Cross-Validation:** Compare at least two configurations using stratified five-fold cross-validation **within the training set**. Vary the representation (e.g., unigrams versus unigrams plus bigrams) or a model setting (e.g., smoothing α or regularization C). Specify your candidates in advance.

3. **Model Selection:** Select by mean positive-class F1. Report each candidate's mean and standard deviation across folds and interpret the comparison.

4. **Preventing Leakage:** Use a pipeline to fit the vectorizer and classifier separately inside each training fold. Vocabulary selection, IDF weights, and feature selection must not use validation-fold or test texts. Refit the selected pipeline on the full training set.

Task 4: Held-Out Evaluation

Freeze your choices, then evaluate on the test set. Do not revise methods or thresholds after inspecting these results.

1. **Performance Table:** Compare (a) a training-set majority-class baseline, (b) the original dictionary, (c) the revised dictionary, and (d) the supervised model in one table. Report accuracy, positive-class precision, recall, and F1. For a training-class tie, the majority baseline predicts negative.

2. **Confusion Matrices:** Plot a confusion matrix for the revised dictionary and for the supervised model. Label the axes and identify which cells represent false positives and false negatives.

3. **Precision-Recall Curves:** Plot their precision-recall curves together, using dictionary scores and model probabilities or decision scores oriented toward the positive class. Add the test set's positive-class prevalence as a horizontal line and mark each method's chosen operating point. Explain when you would favour precision or recall.

Task 5: Interpretation

- For each final method, show two false positives and two false negatives: text, reference label, prediction, score, and a brief explanation. If fewer exist, show all available errors.

- Display the model's ten strongest positive and ten strongest negative word features. Use signed coefficients or, for Naive Bayes, $\log P(w \mid +) - \log P(w \mid -)$. Compare them with your dictionary.

- In 200–300 words, answer the assignment question. Discuss where the dictionary and model differ, the label limitation, and whether these results justify applying either method to political speeches or another new domain.

Submission

Submit an executed Jupyter notebook or rendered R Markdown/Quarto document with runnable source, plots, and written answers.